

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



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Tesi di Laurea Magistrale

The ideally exact categories

A non-pointed generalization
of the notion of semiabelian category

Relatore

professor Alan Cigoli

Candidato

Christian D'Ascoli

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Introduction

Chapter 1

Preliminaries

In order to understand the notions introduced in this thesis is important to recall some definitions and constructions. We also take this opportunity to introduce and fix some notations.

First we define some categories that will be used throughout the thesis. Let $\mathcal{A}$ be a category, we defines the following categories.

- The **slice category**: given an object B in a category $\mathcal{A}$, we call $(\mathcal{A} \downarrow B)$ the category of objects over E , we denote an object by $(A, \alpha : A \rightarrow B)$. A morphism between two objects (A, α) and (A', α') is a morphism $f : A \rightarrow A'$ such that $\alpha' f = \alpha$.

$$\begin{array}{ccc} A & & A \xrightarrow{f} A' \\ \downarrow \alpha & & \downarrow \alpha \swarrow \alpha' \\ B & & B \end{array}$$

- The **coslice category**: given an object B in a category $\mathcal{A}$, we call $(B \downarrow \mathcal{A})$ the category of objects under B , we denote an object by $(\alpha : B \rightarrow A, A)$. A morphism between two objects (α, A) and (α', A') is a morphism $f : A \rightarrow A'$ such that $f\alpha = \alpha'$.

$$\begin{array}{ccc} B & & B \\ \downarrow \alpha & & \downarrow \alpha \searrow \alpha' \\ A & & A \xrightarrow{f} A' \end{array}$$

- The **category of points**: given an object B in a category $\mathcal{A}$, we call $\text{Pt}(B)$ the category of points of B , we denote an object by (A, B, p, s) which is a diagram of the type $A \begin{smallmatrix} \xrightarrow{p} \\ \xleftarrow{s} \end{smallmatrix} B$ with the condition $ps = 1_B$. Morphisms between two objects (A, B, p, s) and (A', B', p', s') are diagrams of the type

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ \swarrow p & & \nwarrow p' \\ & B & \\ \searrow s & & \swarrow s' \end{array}$$

with the conditions $p'f = p$ and $fs = s'$

We recall the following definitions.

Definition 1.0.1. A finitely complete category $\mathcal{A}$ is said to be **regular** if

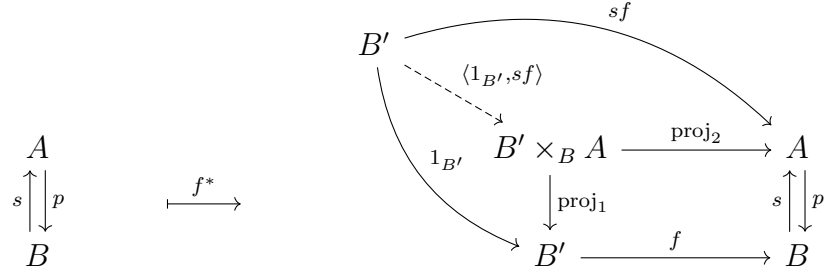
- Every morphism in $\mathcal{A}$ admits an (regular epi, mono) factorization;
- Regular epis in $\mathcal{A}$ are pullback stable.

Definition 1.0.2. A category $\mathcal{A}$ is said to be **exact** if

- It is regular;
- Any equivalence relation is the kernel pair of some arrow.

In order to define the next notion, we need the following construction. Suppose $\mathcal{A}$ has pullbacks and consider a morphism $f : B' \rightarrow B$ in the same category. We define the **change-of-base functor** of f .

$$f^* : \text{Pt}(B) \rightarrow \text{Pt}(B')$$



where $\langle 1_{B'}, sf \rangle$ is the section of proj_1 . It holds that $(\text{proj}_1)\langle 1_{B'}, sf \rangle = 1_{B'}$. The action on a morphism $\alpha : A \rightarrow A'$ is given by



Definition 1.0.3. A finitely complete category $\mathcal{A}$ is said to be **protomodular** if for every morphism $f : A \rightarrow B$ in $\mathcal{A}$, the change-of-base functor $f^* : \text{Pt}(B) \rightarrow \text{Pt}(A)$ is conservative.

We conclude recalling the following.

Definition 1.0.4. A category $\mathcal{A}$ is said to be **semiabelian** if

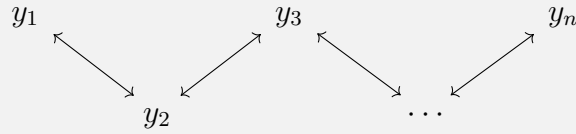
- It is pointed;
- It is exact;
- It is protomodular;
- It has binary coproducts.

Chapter 2

Theory

2.1 Connected limits

Definition 2.1.1. A **connected limit** is a limit over a connected category. A category is connected if it is inhabited and any two objects can be connected by a zigzag of morphisms



where each symbol $\leftrightarrow$ is a morphism $y_i \rightarrow y_{i+1}$ or $y_{i+1} \rightarrow y_i$.

Example 2.1.2. Some examples

- The pullback is a connected limit.

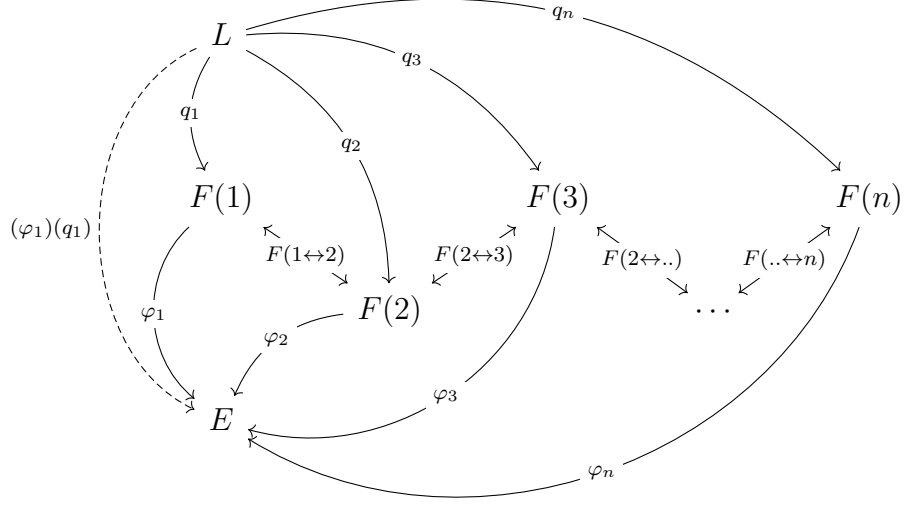
$$\begin{array}{ccc}
 & & 2 \\
 & & \downarrow j \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

$$1 \begin{array}{c} \xrightarrow{i} \\ \xrightarrow{j} \end{array} 2$$

Remark 2.1.3. The connected limits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. In fact we can provide to the object L of the limit cone $(L, (q_i)_{i \in \mathcal{I}})$ in $\mathcal{A}$ a morphism to form an object in $(\mathcal{A} \downarrow E)$. In the zigzag below we chose $(\varphi_1)(q_1)$,

but we could use whatever morphism of the type $(\varphi_i)(q_i)$. The connection guarantees that all the morphisms $q_1 \dots q_n$ involved in the limit in $\mathcal{A}$ are also morphisms in $(\mathcal{A} \downarrow E)$, ie $(\varphi_i)(q_i) = (\varphi_1)(q_1) \forall i$.



By the previous result, the following holds.

Proposition 2.1.4. *Let E be an object in $\mathcal{A}$. Then the forgetful functor $U : (\mathcal{A} \downarrow E) \rightarrow \mathcal{A}$ creates all connected limits that $\mathcal{A}$ has.*

Note that this result is not true for all the limits. This is extensively shown in the following example.

Example 2.1.5. *The product is not a connected limit, in fact as we can observe the following category is not connected*

1 2

We try to compute the product directly in $(\mathcal{A} \downarrow E)$. Let (D, δ) and (D', δ') be two objects in $(\mathcal{A} \downarrow E)$, the product is given by the following pullback

$$\begin{array}{ccccc}
 & D \times_B D' & \longrightarrow & D' & \\
 & \swarrow & & \searrow \delta' & \\
 D & \xrightarrow{\delta} & E & & \\
 & \searrow \delta & \downarrow 1_E & \swarrow \delta' & \\
 & & E & &
 \end{array}$$

which doesn't correspond to the product $D \times D'$ in $\mathcal{A}$.

2.2 The pullback functor and its adjunction

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow B$ in $\mathcal{A}$, one can consider functors pullback and shrink

$$\begin{array}{ccc}
 & \xrightarrow{p^*} & \\
 (\mathcal{A} \downarrow B) & & (\mathcal{A} \downarrow E) \\
 & \xleftarrow{p_!} &
 \end{array}$$

defined as follows:

- $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the **pullback functor**

$$\begin{array}{ccc}
 \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \nearrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & \\ & & B & & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}_1' & \\ E & & & & & \end{array} \\ \text{dashed arc: } p^*(f) = \tilde{f} \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the **shrink functor**

$$\begin{array}{ccc}
 \begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} D & \xrightarrow{g} & D' \\ \delta \downarrow & \nearrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

Proposition 2.2.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then $p_!$ preserves connected limits.*

Proof. By Remark 2.1.3 connected limits are computed in $\mathcal{A}$ and $p_!$ doesn't affect objects and morphism in $\mathcal{A}$. $\square$

Proposition 2.2.2. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then*

$$(\mathcal{A} \downarrow B) \begin{array}{c} \xrightarrow{p^*} \\ \xleftarrow{p_!} \end{array} (\mathcal{A} \downarrow E) \quad \text{with } p_! \dashv p^*$$

is an adjunction. The unit is given by

$$\eta : \text{Id}_{(\mathcal{A} \downarrow E)} \rightarrow p^* p_!$$

$$\eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)$$

and the counit is given by

$$\epsilon : p_! p^* \rightarrow \text{Id}_{(\mathcal{A} \downarrow B)}$$

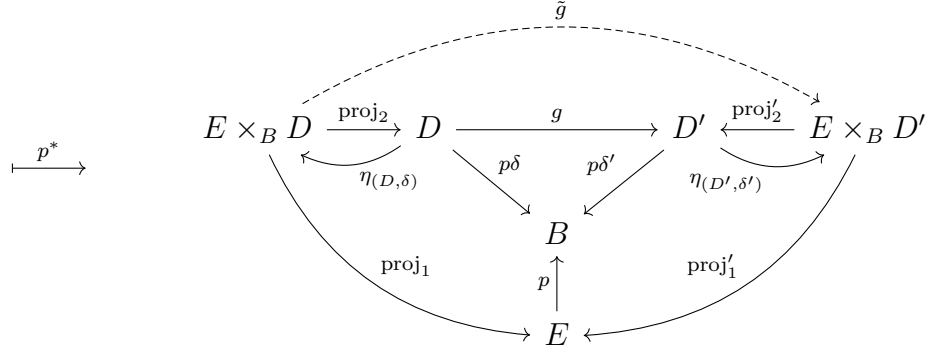
$$\epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)$$

Proof. We want to prove this is adjunction. We start proving that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc} D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc} D & \xrightarrow{g} & D' \\ \downarrow \delta & \searrow \delta' & \\ E & & \end{array} \quad \xrightarrow{p_!} \quad \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ & & B \end{array}$$



Consider the following commutative diagram.

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow & \downarrow p^*p_!(g)=\tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

Since $(\text{proj}'_1, \text{proj}'_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(g) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. Thus the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & A \\
 p_!p^*(f)=\tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

As similarly done above, we need to expand the action of the functor $p_!p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \tilde{f} & & \\ & & \curvearrowright & & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \swarrow \alpha' & & \\ & B & & & \\ \text{proj}_1 \swarrow & & \searrow \text{proj}_1' & & \\ & E & & & \end{array} \end{array} \\
 \\
 \xrightarrow{p!} & & \begin{array}{c} \begin{array}{ccccc} & & \tilde{f} & & \\ & & \curvearrowright & & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \swarrow \alpha' & & \\ & B & & & \\ p(\text{proj}_1) \swarrow & & \searrow p(\text{proj}_1') & & \\ & E & & & \end{array} \end{array}
 \end{array}$$

Since $\epsilon_{(A,\alpha)} = \text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. Thus also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A,α) be an object in $(\mathcal{A} \downarrow B)$.

$$((p^* \epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^* \epsilon)_{(A,\alpha)} (\eta p^*)_{(A,\alpha)} = p^* (\text{proj}_{2,E \times_B A}) (\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A \\ & \searrow \alpha(\text{proj}_{2,E \times_B A}) & \swarrow \alpha \\ & B & \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \text{proj}_{2,\tilde{E} \times_B A} & & \\ & & \curvearrowright & & \\ E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A & \xleftarrow{\text{proj}_{2,E \times_B A}} & E \times_B A \\ & \searrow (\alpha)(\text{proj}_{2,E \times_B A}) & \swarrow \alpha & & \\ & B & & & \\ \text{proj}_{1,E \times_B (E \times_B A)} \swarrow & & \searrow \text{proj}_{1,E \times_B A} & & \\ & E & & & \end{array} \end{array}
 \end{array}$$

$$\circledast = (\text{proj}_{2, E \times_B A}) \langle \eta_{(E \times_B A, \text{proj}_1)} \rangle$$

and then considering the definition of η , we obtain

$$\begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle & & & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow & & \downarrow p(\text{proj}_{1, E \times_B A}) & \\
 \text{proj}_{1, E \times_B A} \searrow & E & \xrightarrow{p} & B & \\
 & & & &
 \end{array}$$

$$\circledast = (\text{proj}_{2, E \times_B A}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.

$$\begin{array}{ccccccc}
 & & 1_{E \times_B A} & & \text{proj}_{2, E \times_B A} & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A & \longleftarrow E \times_B A \\
 & \nearrow \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle & & \searrow (\alpha)(\text{proj}_{2, E \times_B A}) & & \swarrow \alpha & \\
 & & & & B & & \\
 & & & & \uparrow p & & \\
 & & & & E & & \\
 & \nwarrow \text{proj}_{1, E \times_B A} & & \nwarrow \text{proj}_{1, E \times_B A} & & \nwarrow \text{proj}_{1, E \times_B A} & \\
 & E \times_B A & & E & & E &
 \end{array}$$

We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $1_{(E \times_B A, \text{proj}_1)}$. Then, since the projections are jointly monic, we have the result. Following the first composition

$$\begin{aligned}
 & (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B A}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle = \\
 & = (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle = \\
 & = (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = (\text{proj}_{2, E \times_B A})(1_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$(\text{proj}_{1, E \times_B A})(\text{proj}_{2, E \times_B A}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle =$$

$$\begin{aligned}
 &= (\text{proj}_{1, E \times_B (E \times_B A)}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle = \text{proj}_{1, E \times_B A} = \\
 &= (\text{proj}_{1, E \times_B A}) (1_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

Hence

$$\circledast = (\text{proj}_{2, E \times_B A}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle = 1_{(E \times_B A, \text{proj}_1)}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned}
 ((\epsilon p_!) \circ (p_! \eta))_{(D, \delta)} &= ((\epsilon p_!)_{(D, \delta)} (p_! \eta))_{(D, \delta)} = (\epsilon_{p_!(D, \delta)}) p_! \langle \delta, 1_D \rangle = \\
 &= (\epsilon_{(D, p\delta)}) p_! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D} \langle \delta, 1_D \rangle = 1_{(D, p\delta)}
 \end{aligned}$$

$$\begin{array}{ccc}
 E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\
 \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\
 E & \xrightarrow{p} & B
 \end{array}$$

So we can conclude that the adjunction holds. $\square$

Remark 2.2.3. p^* is a right adjoint functor, so it preserves limits. $p_!$ is a left adjoint functor, so it preserves colimits.

Property 2.2.4. The diagram constructed by p^* acting on a morphism $f : (A, \alpha) \rightarrow (A', \alpha')$ is a pullback in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' \\
 \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 \\
 A & \xrightarrow{f} & A'
 \end{array}$$

Proof. We prove it in $\mathcal{A}$, then by *Remark 2.1.3* it holds in $\mathcal{A} \downarrow B$. Since the right and the total diagrams below are pullback, then the left diagram is a

pullback.

$$\begin{array}{ccccc}
 & & \text{proj}_1 & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' & \xrightarrow{\text{proj}'_1} & E \\
 \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 & & \downarrow p \\
 A & \xrightarrow{f} & A' & \xrightarrow{\alpha'} & B \\
 & \searrow & & \nearrow & \\
 & & \alpha & &
 \end{array}$$

□

Property 2.2.5. *The unit η and the counit ϵ are cartesian.*

Proof. First we prove that η is cartesian. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Applying *Property 2.2.4* to $p_!(g)$, we obtain that the following diagram is a pullback.

$$\begin{array}{ccc}
 E \times_B D & \xrightarrow{\text{proj}_2} & D \\
 \downarrow p^*p_!(g)=\tilde{g} & & \downarrow g \\
 E \times_B D' & \xrightarrow{\text{proj}'_2} & D'
 \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc}
 & & 1_D & & \\
 & \nearrow & & \searrow & \\
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\
 \downarrow g & & \downarrow p^*p_!(g)=\tilde{g} & & \downarrow g \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & \searrow & & \nearrow & \\
 & & 1_{D'} & &
 \end{array}$$

Now we want to prove that ϵ is cartesian. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $(\mathcal{A} \downarrow B)$. By *Proposition 2.2.1* applied at the pullback in *Property 2.2.4* for f we obtain that the following diagram is a pullback.

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}=\text{proj}_2} & A \\
 \downarrow p_!p^*(f)=\tilde{f} & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}=\text{proj}'_2} & A'
 \end{array}$$

□

Proposition 2.2.6. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then p^* preserves coequalizer of kernel pairs whose coequalizers are pullback stable regular epimorphisms.*

Proof. Consider the following diagram:

$$\begin{array}{ccccc}
 p^*(\text{Eq}(q)) & \xrightleftharpoons[p_1]{\tilde{p}_2} & p^*(A) = A \times_B E & \xrightarrow{\tilde{q}} & p^*(\text{Coeq}(p_1, p_2)) \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Eq}(q) & \xrightleftharpoons[p_1]{p_2} & A & \xrightarrow{q} & \text{Coeq}(p_1, p_2)
 \end{array}$$

Let q be the coequalizer of the kernel pair $\text{Eq}(q)$. We can affirm by *Property 2.2.4* applied to q that the diagram in the bottom-right corner is a pullback. Since, by hypothesis, q is a pullback stable regular epimorphism, we have that $\tilde{q}$ is a regular epimorphism.

We now aim to show that $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. Since $\tilde{q}$, $\tilde{p}_1$ and $\tilde{p}_2$ are obtained applying p^* to the kernel pair $\text{Eq}(q)$, and p^* preserves limits, it follows that $p^*(\text{Eq}(q))$ is the kernel pair of $\tilde{q}$. We already know that $\tilde{q}$ is a regular epi. Thus, by *Proposition A.0.7*, $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. □

Later on it will be useful to study monads. Thus, we state the following proposition.

Property 2.2.7. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. The associated monad $(p^*p_!, \eta, p^*\epsilon p_!)$ is cartesian.*

Proof. We need to prove that the category $(\mathcal{A} \downarrow E)$ has pullback, the monad preserves pullbacks, and both the multiplication $p^*\epsilon p_!$ and the unit η are cartesian.

The category $(\mathcal{A} \downarrow E)$ has pullbacks, by *Proposition 2.1.4*, since $\mathcal{A}$ has. The monad $p^*p_!$ preserves pullbacks, in fact p^* is a right adjoint so it preserves limits, therefore preserves pullbacks, and $p_!$ preserves pullbacks by *Proposition 2.2.1*. We already proved that the unit η is cartesian, so it remains to prove that the multiplication $p^*\epsilon p_!$ is cartesian. Since ϵ is cartesian and p^* preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. □

2.3 Nullary and essentially nullary monads

Definition 2.3.1. A monad (T, η, μ) on a category $\mathcal{X}$ with finite coproducts is said to be **nullary** if, for each object X in $\mathcal{X}$, the morphism $[T(!_X), \eta_X]$ is an iso, where $!_X$ is the unique morphism from the initial object of $\mathcal{X}$ to X . A monad is said to be **essentially nullary** if, for each object X in $\mathcal{X}$, the morphism $[T(!_X), \eta_X]$ is a strong epi.

Now we claim a lemma which will be useful to prove some theorems.

Lemma 2.3.2. If a category $\mathcal{X}$ is protomodular, then for each commutative diagram of the form

$$\begin{array}{ccc} V & \xrightarrow{u} & X \\ q \downarrow \uparrow t & & p \downarrow \uparrow s \\ W & \xrightarrow{v} & Y \end{array}$$

where p, q are split epimorphisms with respective sections s, t and the downward directed square is a pullback, it holds that the pair (u, s) is a strongly epimorphic pair.

Proof. Assume $\mathcal{X}$ is protomodular. We want to prove that if u and s factorize through a mono m as $u = mf$ and $s = ms'$, then m is an iso. By a standard result, proved in *Proposition A.0.5*, this implies that (u, s) is a strongly epimorphic pair.

Define $p' = pm$, we have that $p's' = 1_Y$, in fact $p's' = pms' = ps = 1_Y$. Pulling back with the change-of-base functor v^* the mono m we obtain the following diagram. In this operation we also generate the morphisms q', t' and w . In the same way as in *Property 2.2.4* we have that the diagram between m and $v^*(m)$ is a pullback.

Now we want to prove that $v^*(m)$ is an iso. Consider the following diagram.

We know that the external diagram commutes by the hypothesis factorization.

$$\begin{array}{ccccc}
 & & & 1_V & \\
 & & & \curvearrowright & \\
 V & \xrightarrow{\langle f, 1_V \rangle} & W & \xrightarrow{v^*(m)} & V \\
 & \searrow f & \downarrow w & & \downarrow u \\
 & & X' & \xrightarrow{m} & X
 \end{array}$$

Since the square is a pullback, it's obvious that $v^*(m)\langle f, 1_V \rangle = 1_V$. It remains to prove that $\langle f, 1_V \rangle v^*(m) = 1_W$.

Notice that $mf v^*(m) = uv^*(m) = mw$, so $fv^*(m) = w$ since m is a mono. Now consider the following compositions, and use what we just noticed.

- $v^*(m)(\langle f, 1_V \rangle v^*(m)) = (v^*(m)\langle f, 1_V \rangle)v^*(m) = 1_V v^*(m) = v^*(m)1_W$
- $w(\langle f, 1_V \rangle v^*(m)) = (w\langle f, 1_V \rangle)v^*(m) = fv^*(m) = w = w1_W$

Since the pair $(v^*(m), w)$ is jointly monic we have that $\langle f, 1_V \rangle v^*(m) = 1_W$. Consdiering this composition and the one above, it follows that $v^*(m)$ is an iso.

By the propomodularity of $\mathcal{X}$, since $v^*(m)$ is an iso, we have that m is an iso, as claimed. $\square$

Theorem 2.3.3. *If $\mathcal{X}$ is a pointed protomodular category with finite coproducts and (T, η, μ) is a monad on $\mathcal{X}$ with cartesian η , then the monad is essentially nullary.*

Proof. For each object X in $\mathcal{X}$ we can consider the following diagram:

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_X} & T(X) \\
 !^X \downarrow \uparrow !_X & & T(!^X) \downarrow \uparrow T(!_X) \\
 0 & \xrightarrow{\eta_0} & T(0)
 \end{array}$$

By hypothesis η is cartesian, so the downward directed square is a pullback. Since $\mathcal{X}$ is protomodular then by *Lemma 2.3.2*, the pair $(\eta_X, T(!_X))$ is strongly epimorphic. By a standard result, proved in *Proposition A.0.6*, this yields that $[\eta_X, T(!_X)]$ is a strong epi. Thus the monad is essentially nullary. $\square$

2.4 Ideally exact categories

Theorem 2.4.1. . *Let $\mathcal{A}$ be a category. The following are equivalent:*

1. $\mathcal{A}$ is exact and protomodular, has finite coproducts and the morphism $! : 0 \rightarrow 1$ is a regular epimorphism.
2. $\mathcal{A}$ is exact, has finite coproducts and there exists $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$.

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\cong} & \mathcal{X}^{UF} \\
 \uparrow F & & \nearrow U \\
 \mathcal{X} & &
 \end{array}$$

3. There exist $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$ and F left adjoint to U such that the associated monad $T = UF$ preserves regular epimorphisms and kernel pairs.

Proof. (1 $\implies$ 3) Suppose $! : 0 \rightarrow 1$ is a regular epi. Since $\mathcal{A}$ is exact, the pullback functor $(!)* : (\mathcal{A} \downarrow 1) \rightarrow (\mathcal{A} \downarrow 0)$ is monadic by (citazione). Notice that $(\mathcal{A} \downarrow 1) = \mathcal{A}$. Now consider $\mathcal{X} = (\mathcal{A} \downarrow 0)$, we want to prove that $\mathcal{X}$ is semiabelian.

$(\mathcal{A} \downarrow 0)$ is pointed, in fact if (A, α) is an object in $(\mathcal{A} \downarrow 0)$, we have the following initial and terminal objects, which are the same:

$$\begin{array}{ccc}
 0 & \xrightarrow{!_A} & A \\
 & \searrow & \downarrow \alpha \\
 & & 0
 \end{array}
 \qquad
 \begin{array}{ccc}
 A & \xrightarrow{\alpha} & 0 \\
 \downarrow \alpha & \nearrow 1_0 & \\
 0 & &
 \end{array}$$

$\mathcal{A}$ has finite coproducts, so $(\mathcal{A} \downarrow 0)$ has finite coproducts. In fact consider $(A, \alpha), (B, \beta)$ in $(\mathcal{A} \downarrow 0)$ and $A + B$ in $\mathcal{A}$, with the canonical injections $\iota_1 : A \rightarrow A + B$ and $\iota_2 : B \rightarrow A + B$. Let (C, γ) be an object in $(\mathcal{A} \downarrow 0)$ with $f : A \rightarrow C$ and $g : B \rightarrow C$ morphisms in $(\mathcal{A} \downarrow 0)$. Providing to $A + B$ the morphism $[\alpha, \beta]$ we have that $(A + B, [\alpha, \beta])$ is the coproduct of (A, α) and (B, β) in $(\mathcal{A} \downarrow 0)$, in fact $\gamma[f, g] = [\alpha, \beta]$, so $[f, g]$ is also a morphism in $(\mathcal{A} \downarrow 0)$.

$$\begin{array}{ccccc}
 A & \xrightarrow{\iota_1} & A + B & \xleftarrow{\iota_2} & B \\
 \searrow \alpha & & \downarrow [\alpha, \beta] & & \swarrow \beta \\
 & & 0 & & \\
 \nearrow [f, g] & & \uparrow \gamma & & \nwarrow g \\
 & & C & &
 \end{array}$$

Now we want to prove that $(\mathcal{A} \downarrow 0)$ is exact, from the exactness of $\mathcal{A}$. First we prove it is finitely complete. It suffices to show that it has terminal a object and pullbacks. The terminal object is the one just shown above. It also has pullbacks, by *Proposition 2.1.4*, since pullbacks are connected limits and $\mathcal{A}$ has them.

Consider now an equivalence relation (R, r_1, r_2) in $(\mathcal{A} \downarrow 0)$, endowed with the morphism ρ and α as shown in the diagram below. We want to prove that such equivalence relation is the kernel pair of some arrow. Since $\mathcal{A}$ is exact, it is a kernel pair in $\mathcal{A}$. By *Proposition A.0.9* we can suppose that it is the kernel pair of its coequalizer.

$$\begin{array}{ccccc} R & \xrightarrow{r_1} & A & \xrightarrow{q} & B \\ & \searrow r_2 & \downarrow \alpha & \swarrow \exists! \beta & \\ & \searrow \rho & 0 & & \end{array}$$

Since r_1 and r_2 are morphism in $(\mathcal{A} \downarrow 0)$, we have that $\alpha r_1 = \rho = \alpha r_2$. Thus, by the universal property of the coequalizer, there exists a unique morphism β such that $\beta q = \alpha$. Providing to B the morphism β we have that (B, β) is the coequalizer of (R, r_1, r_2) in $(\mathcal{A} \downarrow 0)$. Notice that β is a morphism in $(\mathcal{A} \downarrow 0)$ by $\beta q = \alpha$.

We proceed by proving that $(\mathcal{A} \downarrow 0)$ is regular. Both the conditions of the definition of regular category are easily extendable from $\mathcal{A}$ to $(\mathcal{A} \downarrow 0)$. In fact since in $\mathcal{A}$ each morphism admits a (regular epi, mono) factorization, then the same holds in $(\mathcal{A} \downarrow 0)$ just providing to the objects a suitable morphism, as shown below.

$$\begin{array}{ccc} \begin{array}{ccc} Q & & \\ \uparrow e & \searrow m & \\ A & \xrightarrow{f} & B \\ \downarrow \alpha & \swarrow \beta & \\ 0 & & \end{array} & \begin{array}{ccc} K & \xrightarrow{g} & A \xrightarrow{e} B \\ \searrow h & \downarrow \alpha & \swarrow \exists! t \\ \alpha g = h g & 0 & \end{array} & \begin{array}{ccc} Q & \xrightarrow{m} & B \\ \downarrow \beta m & \swarrow \beta & \\ 0 & & \end{array} \end{array}$$

In order to prove that regular epis are pullback stable in $(\mathcal{A} \downarrow 0)$ we can use the fact that they are in $\mathcal{A}$. In fact, having a regular epi in $\mathcal{A}$ we can proceed providing a suitable morphism to the coequalizer as done above.

We must still establish that the $(\mathcal{A} \downarrow 0)$ is protomodular. Since the construction of the change-of-base functor involves pullbacks, which are a connected limits, then by *Remark 2.1.3* the construction is the same in $(\mathcal{A} \downarrow 0)$ and in $\mathcal{A}$. Thus $(\mathcal{A} \downarrow 0)$ inherits protomodularity from $\mathcal{A}$, noting that the definition of iso in $\mathcal{A}$ and $(\mathcal{A} \downarrow 0)$ is the same. Hence, since $\mathcal{X} = (\mathcal{A} \downarrow 0)$ is exact, protomodular, pointed and has binary coproducts, we can conclude that it is semiabelian.

It remains to prove that the monad $(!)*(!)_!$ preserves regular epimorphisms and kernel pairs. Consider a regular epi $q : A \rightarrow B$ in $(\mathcal{A} \downarrow 0)$, it is the coequalizer of some parallel arrows. Since the coequalizer is a colimit, it is preserved by the left adjoint functor $(!)_!$. So it suffices to prove that, with $\mathcal{A}$

regular, $(!)^*$ preserves regular epis. Consider the following diagram:

$$\begin{array}{ccc}
 0 \times_1 A & \xrightarrow{(!)^*(q)} & 0 \times_1 B \\
 \downarrow & & \downarrow \\
 A & \xrightarrow{q} & B \\
 & \searrow & \swarrow \\
 & 1 &
 \end{array}$$

By *Property 2.2.4* the square is a pullback. Since $\mathcal{A}$ is regular, and hence so is $(\mathcal{A} \downarrow 0)$, it follows that $(!)^*$ is a regular epi due to the stability of regular epis under pullbacks.

Finally, we want to establish that the monad $(!)^*(!)$ preserves kernel pairs. This is obvious: $(!)$ preserves connected limits, such as kernel pairs, by *Proposition 2.2.1*, while $(!)^*$ preserves kernel pairs since it is a right adjoint. Thus $(!)^*(!)$ preserves kernel pairs.

(3 $\implies$ 2) Suppose there exists $\mathcal{X}$ semiabelian category and a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$ satisfying the stated conditions. First we need to verify that $\mathcal{A}$ is exact. $\square$

Definition 2.4.2. A category $\mathcal{A}$ is said to be **ideally exact** if it satisfies one of the equivalent conditions of Theorem 2.4.1.

Appendix

Definition A.0.1. Let γ be a natural transformation between two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, we say that γ is **cartesian** if for every morphism $f : A \rightarrow B$ in $\mathcal{C}$, the following diagram is a pullback:

$$\begin{array}{ccc} F(A) & \xrightarrow{\gamma_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\gamma_B} & G(B) \end{array}$$

ie the naturality diagrams are pullbacks.

Definition A.0.2. Let $\mathcal{C}$ be a category and $f : A \rightarrow B$. We say that f is a **strong epimorphism** if it for each commutative diagram of solid arrows

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ u \downarrow & \exists! d \swarrow & \downarrow v \\ C & \xrightarrow{m} & D \end{array}$$

where m is a mono, there exists a unique morphism d such that $df = u$ and $md = v$.

Proposition A.0.3. Let $\mathcal{C}$ be a category with pullbacks and $f : A \rightarrow B$. The following are equivalent:

1. f is a strong epimorphism;
2. Given a factorization $f = mg$, if m is a mono, then it is an iso.

Proof. (1 $\implies$ 2) Let $f = mg$ be a factorization with m mono and consider

the square:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & \searrow d & \downarrow 1_B \\ C & \xrightarrow{m} & B \end{array}$$

By (1) we have $md = 1_B$. Furthermore $mdm = m$ and, since m is mono, we have that $dm = 1_C$. Thus m is an iso.

(2 $\implies$ 1) We want to prove that f is a strong epi. Consider the pullback

$$\begin{array}{ccccc} A & & & & \\ & \searrow & & \searrow f & \\ & & C \times_D B & \xrightarrow{\text{proj}_2} & B \\ & \swarrow u & \downarrow \text{proj}_1 & & \downarrow v \\ & & C & \xrightarrow{m} & D \end{array}$$

Since m is a mono, the projection proj_2 is a mono. Now f factors through proj_2 , so by (2) proj_2 is an iso. We define $d = \text{proj}_1(\text{proj}_2)^{-1}$, which is the morphism we are looking for. $\square$

Definition A.0.4. Let $\mathcal{C}$ be a category and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. We say that the pair (f, g) is a **strongly epimorphic pair** if for each commutative diagram of solid arrows

$$\begin{array}{ccccc} & & B & & \\ & g \nearrow & \downarrow f & \searrow h & \\ A' & & A & & D \\ & g' \searrow & \downarrow f' & \nearrow m & \\ & & C & & \end{array}$$

$\exists! d$

where m is a mono, there exists a unique morphism d such that $f' = df$, $g' = dg$ and $md = h$.

Proposition A.0.5. Let $\mathcal{C}$ be a category with pullbacks and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. The following are equivalent:

1. (f, g) is a strong epimorphic pair;
2. Given factorizations $f = mf'$ and $g = mg'$, if m is a mono, then it is an iso.

Proof. By adding the morphism g' and its conditions in the proof of *Proposition A.0.3*, we obtain the proof of the current proposition. $\square$

Proposition A.0.6. *Let $\mathcal{C}$ be a category with binary coproducts and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. The following are equivalent:*

1. (f, g) is a strongly epimorphic pair;
2. $[f, g]$ is a strong epimorphism.

Proof. (1 $\implies$ 2) (f, g) is a strongly epimorphic pair. Consider the commutative diagram where m is a mono

$$\begin{array}{ccc} A + A' & \xrightarrow{[f, g]} & B \\ \downarrow l & & \downarrow h \\ C & \xrightarrow{m} & D \end{array}$$

Composing with the injection $\iota_1 : A \rightarrow A + A'$ and $\iota_2 : A' \rightarrow A + A'$, we have the following commutative diagram

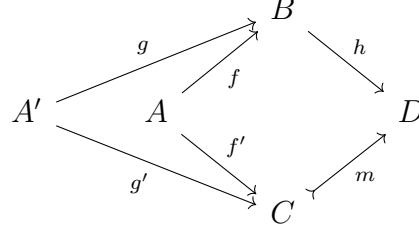
$$\begin{array}{ccccc} & & B & & \\ & g \nearrow & & h \searrow & \\ A' & & A & & D \\ & \nwarrow l_2 & \nwarrow l_1 & & \\ & & C & & \end{array}$$

Since (f, g) is strongly epimorphic, there exists a unique morphism $d : B \rightarrow C$ such that $l_1 = df$, $l_2 = dg$ and $md = h$. The first two yield the equalities $d[f, g]\iota_1 = df = l_1$ and $[f, g]\iota_2 = dg = l_2$. So, by the universal property of the coproduct, we have that $d[f, g] = l$.

$$\begin{array}{ccc} A & & A \\ \downarrow \iota_1 & & \downarrow \iota_1 \\ A + A' & \xrightarrow{[f, g]} B \xrightarrow{d} C & A + A' \xrightarrow{l} C \\ \uparrow \iota_2 & & \uparrow \iota_2 \\ A' & & A' \end{array}$$

We already know that $md = h$, hence $[f, g]$ is a strong epimorphism.

(2 $\implies$ 1) $[f, g]$ is a strong epimorphism. Consider the following commutative diagram where m is a mono



Thus we have the commutative diagram

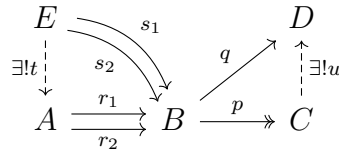
$$\begin{array}{ccc} A + A' & \xrightarrow{[f, g]} & B \\ \downarrow [f', g'] & & \downarrow h \\ C & \xrightarrow{m} & D \end{array}$$

Since $[f, g]$ is a strong epimorphism, there exists a unique morphism $d : B \rightarrow C$ such that $d[f, g] = [f', g']$ and $md = h$. Hence $f' = [f', g']\iota_1 = d[f, g]\iota_1 = df$ and $g' = [f', g']\iota_2 = d[f, g]\iota_2 = dg$. Thus (f, g) is a strongly epimorphic pair. $\square$

Proposition A.0.7. *If a regular epi has a kernel pair, then it is the coequalizer of its kernel pair.*

Proof. Let p be a regular epi, and let (r_1, r_2) be the kernel pair of p . Since p is a regular epi, it is the coequalizer of some parallel arrows. Let (s_1, s_2) be such pair as in the diagram below. We want to prove that p is the coequalizer of (r_1, r_2) .

We already know that $pr_1 = pr_2$, as (r_1, r_2) is the kernel pair of p . It remains to establish the universal property of the coequalizer. Let $q : B \rightarrow D$ be a morphism such that $qr_1 = qr_2$. By the universal property of the kernel pair, there exists a unique morphism $t : E \rightarrow A$ such that $r_1t = s_1$ and $r_2t = s_2$. We have that $qs_1 = qr_1t = qr_2t = qs_2$, so, by the universal property of the coequalizer for the pair (s_1, s_2) , there exists a unique morphism $u : C \rightarrow D$ such that $q = up$. Thus the universal property of the coequalizer also holds for the pair (r_1, r_2) .



$\square$

Proposition A.0.8. *If a kernel pair has a coequalizer, then it is the kernel pair of its coequalizer.*

Proof. Let (K, r_1, r_2) be a kernel pair of some arrow $q : A \rightarrow B$, and assume that such kernel pair has a coequalizer $e : B \rightarrow Q$. We want to prove that (r_1, r_2) is the kernel pair of its coequalizer e .

$$\begin{array}{ccccc} & & & & Q \\ & & & \nearrow e & \\ K & \xrightarrow[r_2]{r_1} & A & \xrightarrow{q} & B \end{array}$$

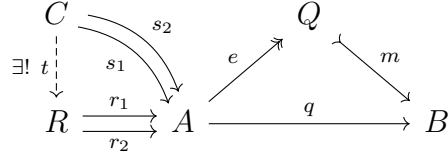
We already know that $er_1 = er_2$, since e is the coequalizer of (r_1, r_2) . It remains to show that the universal property of the kernel pair holds. Consider $h : X \rightarrow A$ and $h' : X \rightarrow A'$ morphisms such that $eh = eh'$. Since (r_1, r_2) is the kernel pair of q , it holds that $qr_1 = qr_2$. Thus by the universal property of the coequalizer e , there exists a unique morphism $l : Q \rightarrow B$ such that $q = le$. Composing with such morphism l we have that $leh = leh'$, hence $qh = qh'$. Since (r_1, r_2) is the kernel pair of q , by the universal property there exists a unique morphism $g : X \rightarrow K$ such that $r_1g = h$ and $r_2g = h'$. Hence (r_1, r_2) is the kernel pair of e , its coequalizer.

□

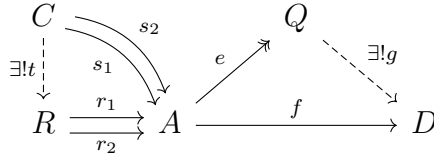
Proposition A.0.9. *In an exact category $\mathcal{A}$, each equivalence relation has a coequalizer and it is the kernel pair of its coequalizer.*

Proof. Consider an equivalence relation (R, r_1, r_2) in $\mathcal{A}$. Since $\mathcal{A}$ is exact, (R, r_1, r_2) is the kernel pair of some arrow $q : A \rightarrow B$. $\mathcal{A}$ is also regular, so q admits a factorization $q = me$, where m is a mono and e is a regular epi, ie the coequalizer of some arrows. Let s_1 and s_2 be such arrows, we have that $es_1 = es_2$. Composing with m , we have that $qs_1 = mes_1 = mes_2 = qs_2$. By the universal property of the kernel pair, there exists a unique morphism

$t : C \rightarrow R$ such that $r_1 t = s_1$ and $r_2 t = s_2$.



Now we want to prove that e is the coequalizer of (r_1, r_2) . We have that $mer_1 = fr_1 = fr_2 = mer_2$. Thus, since m is a mono, $er_1 = er_2$. It remains to prove the universal property of the coequalizer. Let $f : A \rightarrow D$ be a morphism such that $fr_1 = fr_2$. Composing with t we have $fr_1 t = fr_2 t$, hence $fs_1 = fs_2$. By the universal property of the coequalizer e , as coequalizer of s_1 and s_2 , there exists a unique morphism $g : Q \rightarrow D$ such that $ge = f$. Hence e is also the coequalizer of r_1 and r_2 .



Finally, we observe that (r_1, r_2) is the kernel pair of e , since by *Proposition A.0.8*, a kernel pair that admits a coequalizer is the kernel pair of its coequalizer.

□